

Q.No:6

DS	Session	Tree 1	Question Information	Level 1	Challenge 66
Question description A company has n employees, who form a tree hierarchy where each employee has a boss, except for the general director. Your task is to process q queries of the form: who is employee x's boss k levels higher up in the hierarchy? Input The first input line has two integers n and q: the number of employees and queries. The employees are numbered $1, 2, \dots, n$, and employee 1 is the general director. The next line has $n-1$ integers $e_2, e_3, \dots, e_n$: for each employee $2, 3, \dots, n$ their boss. Finally, there are q lines describing the queries. Each line has two integers x and k: who is employee x's boss k levels higher up? Output Print the answer for each query. If such a boss does not exist, print -1. Constraints $1 \leq n, q \leq 10^5$ $1 \leq e_i \leq i-1$ $1 \leq x \leq n$ $1 \leq k \leq n$					
Logical Test Cases Test Case 1 INPUT (STDIN) <pre>5 3 1 3 2 2 4 2 1 2 1 3</pre> EXPECTED OUTPUT <pre>1 -1 -1</pre> Test Case 2 INPUT (STDIN) <pre>5 3 1 1 3 3 4 1 4 2 4 3</pre> EXPECTED OUTPUT <pre>3 1 -1</pre> Mandatory Test Cases Test Case 1 KEYWORD <pre>void link(int i, int j)</pre>					

Code:

```
#include <stdio.h>

int main()
{
    int n, q;
    int boss[100005];

    scanf("%d %d", &n, &q);

    boss[1] = 0;

    for (int i = 2; i <= n; i++)
    {
        scanf("%d", &boss[i]);
    }

    int x, k;
    int queryX[100005];
    int queryK[100005];

    for (int i = 0; i < q; i++)
    {
```

```
scanf("%d %d", &queryX[i], &queryK[i]);
}
printf("Output:\n");
for (int i = 0; i < q; i++)
{
    x = queryX[i];
    k = queryK[i];

    while (k > 0 && x != 0)
    {
        x = boss[x];
        k--;
    }

    if (x == 0)
        printf("-1\n");
    else
        printf("%d\n", x);
}

return 0;
}
```

Output:

Test_case 1:

```
5 3
1 3 2 2
4 2
1 2
1 3
Output:
1
-1
-1
```

Test_case 2:

```
5 3
1 1 3 3
4 1
4 2
4 3
Output:
3
1
-1
```